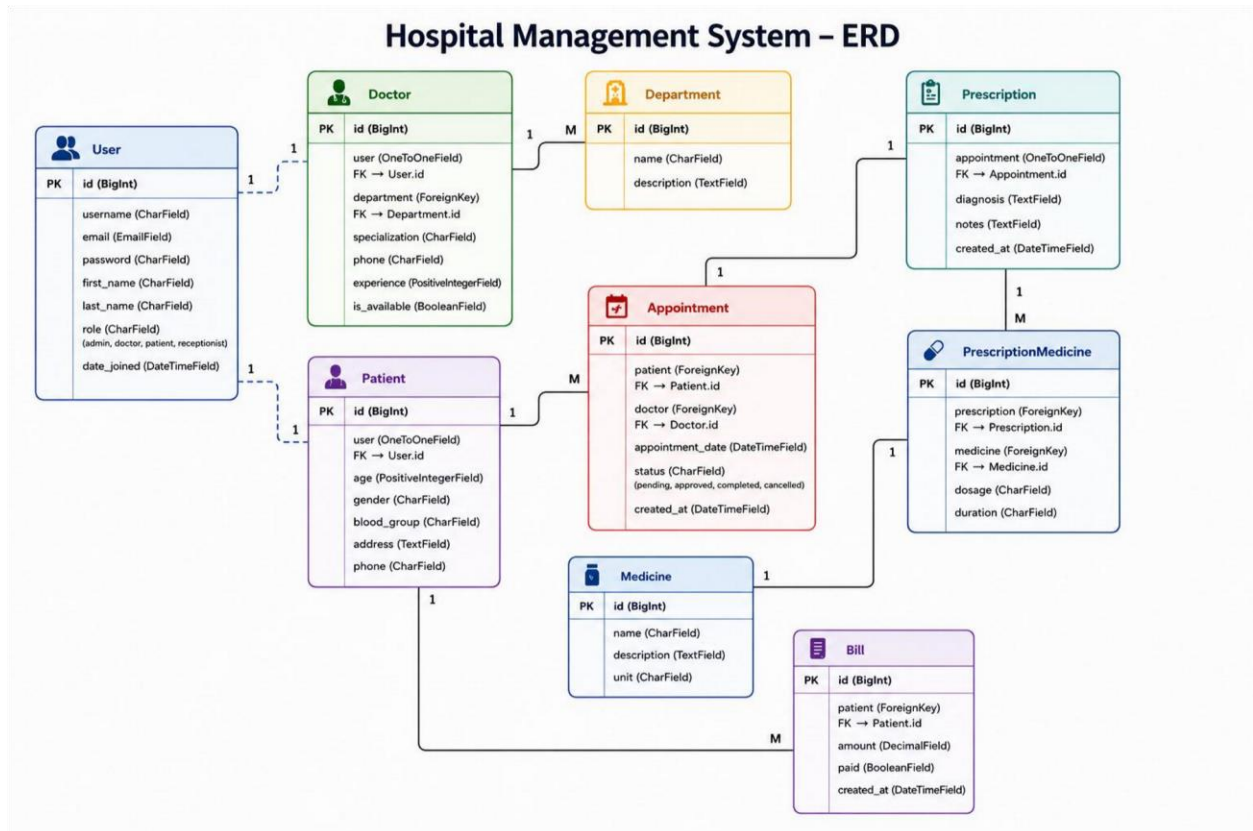


## Assignment: Hospital Management System



For the Hospital Management System assignment you're working on in Module 24 using Django REST Framework, complete the **Frontend part** (based on the ERD and requirements below).

**Objective:** Develop a RESTful API for a Hospital Management System using Django REST Framework (DRF) based on the provided ERD, along with a React.js frontend that consumes these APIs.

### Requirements

#### 1. Backend Development (DRF)

- Implement the database exactly as per the given ERD.

- Create Django Models for all entities with proper fields and relationships.
- Develop REST APIs using Django REST Framework.

## **2. Models (Must Follow ERD)**

You must implement these models exactly:

- User
- Doctor
- Patient
- Department
- Appointment
- Prescription
- PrescriptionMedicine
- Medicine
- Bill

## **3. Core API Features**

Implement the following APIs with proper authentication and permissions:

1. User Management
  - Register / Login (different roles: admin, doctor, patient, receptionist)
2. Doctor & Patient APIs
  - CRUD for Doctors and Patients
  - Doctor availability
3. Appointment APIs
  - Book, View, Update, Cancel Appointment
  - Filter by doctor/patient/date
4. Prescription APIs
  - Create prescription with multiple medicines
  - View prescriptions
5. Medicine APIs
  - List / Search medicines

## 6. Billing APIs

- Generate bill
- Mark as paid

## 4. Frontend Development (React)

Develop a React.js application that connects to the DRF APIs above and provides a working UI for all core features.

- Role-based Login/Register pages (admin, doctor, patient, receptionist), with JWT/Token stored and attached to API requests.
- Protected routes so each role only accesses the pages relevant to it.
- Doctor & Patient management: list views, CRUD forms, doctor availability toggle.
- Appointment module: booking form, view/update/cancel actions, filter by doctor/patient/date.
- Prescription module: create a prescription with multiple medicines (dynamic add/remove fields), view prescription history.
- Medicine module: searchable/listable medicine table.
- Billing module: generate bill for a patient, mark bill as paid, show paid/unpaid status.
- Use Axios (or Fetch) for API calls, with proper loading and error states.

## Technical Requirements

- Django + Django REST Framework
- Use PostgreSQL (preferred) or SQLite
- Token Authentication / JWT
- Proper serializers, viewsets, and routers
- Role-based permissions
- Input validation & error handling
- React.js frontend (with React Router, Axios) consuming the above APIs

## Submission

- Submit the GitHub repository link (backend and frontend, either in one repo with backend/ and frontend/ folders, or as separate repos linked in the README)